

# Migraine: 1. diagnosis in adults

SIGN 155, 2023, NEJM 2017;377:553, Lancet 2018;391:1315, BJGP 2020;70:88, NICE 2012 (updated 2025), CG150, NICE 2019, NG127, NEJM 2020;383:1866, BMJ 2022;379:e067670



# Red Whale

GEMS

Guidelines & Evidence Made Simple

## Could this be migraine?

Rule out red flags (see GEMS on headache diagnosis).

Look for the following features, remembering that:

- Migraine can occur with or without aura.
- Headache doesn't necessarily need to be present; some people get migraine aura without headache.

### Premonitory symptoms (occur hours–days before headache)

Yawning  
Light sensitivity  
Nausea  
Diarrhoea  
Food cravings  
Increased urination  
Neck/sinus pain  
Impaired concentration  
Fatigue  
Mood changes, e.g. anxiety, depression, irritability

### Aura (if it occurs)

Classically lasts 5–60min.  
Usually precedes headache.  
More info below.

### Headache and associated symptoms

**Headache** (average time to maximum intensity 1h, average duration 24h):

- Unilateral/throbbing.
- Worse with activity.
- Can change sides during attack/with next episode.

#### Associated symptoms:

- Photophobia.
- Phonophobia.
- Allodynia (pain related to touch).
- Nausea/vomiting.

### Postdromal symptoms

Time between headache finishing and feeling better. Occurs in 80% of people, usually lasts <12h.

- Symptoms similar to premonitory symptoms.
- Can get short-lived head pain with sudden head movement.

- With aura, symptoms may be:
  - POSITIVE (developing a new symptom), e.g. flashing lights, tingling sensations.
  - NEGATIVE (loss of something), e.g. visual loss, numbness.
- Sensory changes are most common in the arm or perioral region. Motor symptoms are rare (e.g. hemiplegic migraine).

### What makes an aura, an aura?

The International Classification of Headache Disorders defines it as follows:

One or more of the following that has fully resolved:  
≥2 episodes of  
Visual  
Sensory  
Speech/language  
Motor  
Brainstem, e.g. vertigo  
Retinal symptoms

PLUS

THREE or more of the following:

- Headache comes with the aura or occurs within 60mins.
- Each aura symptom lasts 5–60min.
- ≥1 aura symptom spreading gradually over ≥5min.
- ≥2 aura symptoms happening in succession.
- ≥1 unilateral aura symptom.
- ≥1 positive aura symptom.

...and no other diagnosis more likely

## Is it an aura or something else?

At times, it may not be clear if this is migraine aura or something else, e.g. TIA, giant cell arteritis, raised intracranial pressure, occipital seizures, postural hypotension, retinal detachment – many of which need same-day referral.

Migraine aura usually gradually evolves, while TIAs often present suddenly, but it is not always clear-cut.

THINK STROKE and treat as CVA/TIA until proven otherwise if **SUDDEN ONSET**:

- Limb or facial weakness.
- Transient unilateral numbness.
- Dizziness with a focal neurological defect.
- Unsteady gait.
- Speech/language disturbance. (NICE NG127)

If aura (+/- headache) and any ONE of the following, consider further investigation/referral (may be alternative diagnosis):

- Motor weakness.
- Double vision.
- Unilateral visual symptoms.
- Poor balance.
- Decreased consciousness. (NICE CG150, 2012)

## Initial and follow-up appointments

### At initial appointment:

- Provide information on migraine.
- Suggest a headache diary.
- Consider stressors/lifestyle factors.
- Offer acute treatment.
- Is prophylaxis needed?

### At follow-ups:

- Revisit symptoms, stressors, impact on quality of life.
- Review headache diary.
- Clarify understanding; provide further resources if needed.
- Consider possibility of medication-overuse headache; address if needed.
- Revisit use of/response to acute/prophylactic treatment.
- Is referral indicated?

(SIGN 155, 2023)

## Migraine: 2. Acute treatment in ADULTS

SIGN 155, 2023, NICE CG150, 2012 (updated 2025), BNF, accessed June 2023, NEJM 2017;377:553, Lancet 2021;397:1505, BMJ 2022;379:e067670



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This GEMS applies to non-pregnant adults.

SIGN's recommendations can be *considered* for adolescents (except aspirin <16y because of the risk of Reye's syndrome).

Treatment choice should consider relevant comorbidities and contraindications e.g. asthma, depression, hypertension, cardiovascular disease, cerebrovascular disease, contraception and risk of pregnancy.

### What do patients need to know about acute treatment?

- **AIM:** to stop/significantly reduce the severity of the migraine (aiming to be pain-free at 2h and remain so for 24h).
- Won't always work for every migraine.
- Triptans:
  - Usually work in 2 out of 3 attacks (this is considered a good treatment response).
  - **Take as soon as headache starts (if migraine with aura: start with headache, not with aura).**
- Warn about the risk of medication-overuse headache, and restrict acute treatment to  $\leq 2$ d/w.
- Encourage adequate hydration with water, regular meals, adequate sleep and exercise, and avoid known triggers, e.g. skipped meals, caffeine intake, alcohol, irregular sleep and stress. Evidence for dietary modification isn't strong.
- Medications that exacerbate migraine: oral contraceptives, HRT, nasal decongestants, SSRIs, PPIs, regular opioids.

### Options for acute treatment

NNTs are for being pain-free after 2h.

Do NOT use opioids or ergots.

If unable to take first-line options, consider:

**Aspirin  
900mg**

NNT = 8  
or

**Sumatriptan  
orally 50–100mg**

NNT 5–6 for oral/nasal  
NNT 2 for subcut  
(zolmitriptan NNT 4.9 (oral) and 3.0 (nasal))

**Metoclopramide  
or prochlorperazine  
10mg**

Can be used for:

- TREATMENT of headache in acute migraine.
- Managing migraine-related nausea/vomiting.

**Do not use metoclopramide regularly: risk of extrapyramidal side-effects.**

**Paracetamol  
1g**

NNT 12

**Ibuprofen 400mg**  
(if ineffective,  
increase to 600mg)

NNT = 7

**If poor response to one triptan, try  
different triptan/different route.**

#### Triptan contraindications:

- Uncontrolled hypertension.
- Cardiovascular disease, peripheral vascular disease or cerebrovascular disease.

**Discontinue if:** heat/heaviness/pressure/tightness anywhere in the body, including throat/chest.

### How to choose/use acute treatments

There are different ways to take acute treatments:

| 'Stepped' treatment                                                                                                                                                                 | 'Stratified' treatment                                                                                                                                                                | Dual therapy                                                                                                                                                                                                                                                                                                                      |  | Monotherapy                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------|
| Initially, try aspirin or ibuprofen. If not successful after 3 headaches, step up to triptan.<br><br><br><b>Aspirin or ibuprofen</b> → <b>Triptan</b><br><i>SIGN recommendation</i> | <b>With each migraine, patient chooses depending on severity.</b><br><br><br><b>Milder</b> → <b>More severe</b><br><b>e.g. aspirin</b> → <b>Triptan</b><br><i>SIGN recommendation</i> | <b>Patient takes combination therapy.</b><br>Recommended by: <ul style="list-style-type: none"><li>• <b>SIGN: for moderate–severe migraines.</b></li><li>• <b>NICE: for all.</b></li></ul><br>SIGN recommends: <b>Triptan (any route) PLUS Aspirin or NSAID</b><br>NICE recommends: <b>Oral triptan PLUS NSAID or paracetamol</b> |  | NICE recommends for those who don't want dual therapy:<br>Aspirin or Paracetamol or NSAID or oral triptan |

### If early or persistent vomiting

- Consider adding antiemetic: **metoclopramide or prochlorperazine.**
- **If using a triptan, consider alternative route:** e.g. nasal or subcutaneous. (Orodispersible triptans are gastrically absorbed, as is much of the nasal triptans.)

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. June 2025. For full references see the relevant Red Whale articles.

## 3. Prophylaxis in NON-PREGNANT ADULTS

SIGN 155, 2023, NICE CG150, 2012 (updated 2025) BNF, accessed May 2023, FSRH CEU statement, Feb 2018, Government advice on valproate, December 2022, Pregnancy Prevention Programme, September 2018



Most of the data on migraine prophylaxis is from those with episodic migraine. SIGN felt that both episodic and chronic migraine are likely on a spectrum, and so those with chronic migraine may also get benefit.

When choosing treatments, take into account comorbidities, contraindications and risk of pregnancy.

## What are we trying to achieve with migraine prophylaxis?

**When to start?** Consider degree of disability, quality-of-life impact, frequency (>1/w) or prolonged, severe attacks.

**Aim is a reduction in migraine headache severity and frequency of 50%:** total control may not be achieved.

**When to review?** Use each treatment for ≥3m at the maximum tolerated dose before deciding if effective or not.

Reassess the need for ongoing prophylaxis after 6–12m.

## Treatment options (see main article for a more in-depth summary of prophylactic medications)

Consider the risk of medication-overuse headache; overuse of acute medication can limit efficacy of prophylaxis. For each treatment option, start at low dose and increase dose based on tolerability and efficacy.

**NICE recommends propranolol, topiramate or amitriptyline as first-line choices, taking into account individual benefits and risks (NICE 2012 (updated 2025), CG150).**

|                 |                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propranolol     | 80–160mg/d, target dose 80mg twice a day. Not in asthma.<br>(Propranolol increases plasma concentration of rizatriptan: maximum dose of rizatriptan 5mg. Do not take within 2h of propranolol.)<br><b>SIGN: FIRST-LINE CHOICE.</b> Consider other options if contraindicated, not tolerated or ineffective.                                                                    |
| Topiramate      | 50–100mg/d, target dose 50mg twice a day (200mg/d no more effective than 100mg/d).<br>( <b>MHRA Pregnancy Prevention Programme: strict requirements for safe prescribing and annual review.</b> Need highly effective contraception: implant does NOT qualify (topiramate is an enzyme inducer)).<br>Common side-effect: depression. Renal excretion: reduce dose if CrCl ≤70. |
| Amitriptyline   | Target dose 30–50mg at night (can be used at a dose of 25–150mg at night).<br>(If sedative effect, can consider an alternative tricyclic antidepressant.)                                                                                                                                                                                                                      |
| Candesartan     | Target dose 16mg once daily. Unlicensed, well tolerated, not recommended in pregnancy (SIGN).                                                                                                                                                                                                                                                                                  |
| Riboflavin (B2) | May help reduced migraine frequency and severity in some patients (400mg once a day) (NICE).                                                                                                                                                                                                                                                                                   |

## Patients at risk of pregnancy

- Topiramate and valproate are teratogenic.
- Candesartan contraindicated in pregnancy.
- Beta-blocker use is heavily cautioned in pregnancy.
- Amitriptyline: only if benefits outweigh risks.

**All who are planning a pregnancy should have migraine prophylaxis reviewed.**

*For details about risks of each drug in pregnancy and who needs highly-effective contraception: see article.*

## Non-drug options

|               |                                                                                                                                                                                                                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acupuncture   | <ul style="list-style-type: none"> <li>• NICE recommends up to 10 sessions over 5–8w if propranolol and topiramate are ineffective/unsuitable.</li> <li>• Reduced frequency of episodic migraine; may be as good as prophylactic drugs (BJGP 2018;68:464).</li> </ul>                                                            |
| CBT           | <ul style="list-style-type: none"> <li>• Some patients report less pain during migraine attacks (BJGP 2018;68:464).</li> <li>• Reduced psychological symptoms (depression, anxiety, sleep disturbance) (BJGP 2018;68:464).</li> <li>• Lancet concluded evidence stronger than for acupuncture (Lancet 2021;397:1505).</li> </ul> |
| Mindfulness   | <ul style="list-style-type: none"> <li>• Reduced use of acute therapies (BJGP 2018;68:464).</li> <li>• If chronic migraine AND coexisting medication-overuse headache, may reduce headache frequency by a similar amount as taking preventative drugs (BJGP 2018;68:464).</li> </ul>                                             |
| Other options | <ul style="list-style-type: none"> <li>• Consider relaxation, including mindfulness, yoga or meditation (SIGN).</li> <li>• Some evidence suggests that massage and biofeedback helps. The following may be beneficial: spinal manipulation; some supplements and botanicals; hydrotherapy (BMJ 2017;357:j1805).</li> </ul>       |

## Referral (consider if ≥3 failed prophylactic treatment options)

Secondary care may consider:

- Flunarizine or CGRP monoclonal antibodies for episodic or chronic migraine.
- Botox A for chronic migraine.
- Sodium valproate: no one (male or female) <55y should be started on valproate unless 2 specialists consider there is no other appropriate treatment option. Follow Valproate Pregnancy Prevention Programme.



Treatment choice should consider relevant comorbidities and contraindications.

### Menstrual migraine (consider if occurs on 2 out of 3 consecutive cycles; use a headache diary)

Oestrogen drops just before menstruation, which can trigger migraine. Migraine before and during menstruation can be more severe/harder to treat. There are two types:

- Pure menstrual migraine: migraine only happens around onset of period.
- Menstrually-related migraine: migraine around onset of period but may occur at other times too.

If not responding to standard acute migraine treatment and has a regular menstrual cycle, prophylactic triptans can be used off licence from 2d before until 3d after bleeding starts (5d total):

|              |                                                                              |
|--------------|------------------------------------------------------------------------------|
| Frovatriptan | 2.5mg twice a day: from 2d before until 3d after bleeding starts (5d total). |
|--------------|------------------------------------------------------------------------------|

Alternatives to frovatriptan:

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| Zolmitriptan | 2.5mg three times a day: from 2d before until 3d after bleeding starts (5d total). |
|--------------|------------------------------------------------------------------------------------|

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| Naratriptan | 2.5mg twice a day: from 2d before until 3d after bleeding starts (5d total). |
|-------------|------------------------------------------------------------------------------|

If also using triptans at other times in the month, the risk of developing medication-overuse headache is increased.

### Pregnancy

#### Prophylaxis:

- Migraine without aura often improves during pregnancy so aim to stop prophylaxis before becoming pregnant.
- Migraine with aura often remains the same in pregnancy:
  - Discuss this when starting migraine prophylaxis if there is the potential to become pregnant.
  - If pregnant, seek specialist advice.

#### Acute treatment:

|                         |                                |
|-------------------------|--------------------------------|
| Paracetamol: first line | Can be used in all trimesters. |
|-------------------------|--------------------------------|

Other treatment options (do not use aspirin for migraine treatment in pregnancy):

|           |                                                    |
|-----------|----------------------------------------------------|
| Ibuprofen | NSAID of choice until 28w. Avoid NSAIDs after 28w. |
|-----------|----------------------------------------------------|

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Triptans | <p>If treatment with paracetamol or ibuprofen fails, a triptan can be considered in all stages of pregnancy (preference for sumatriptan). The BNF says limited evidence: avoid unless potential benefit outweighs the risks. Risks should be discussed:</p> <ul style="list-style-type: none"> <li>• Not associated with increased risk of premature delivery or major congenital malformation (best evidence for sumatriptan).</li> <li>• Spontaneous abortion: risk increased in one meta-analysis, not in a larger cohort study.</li> <li>• Possible increased risk of externalising behaviour problems in the child (e.g. hitting) (cohort study, low-quality evidence).</li> </ul> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Other causes of headache in pregnancy:

Other, life-threatening, causes of headache in pregnancy include pre-eclampsia or cerebral venous thrombosis. We should be mindful of these differentials, carry out a full neurological examination (including fundoscopy) and check blood pressure and urinalysis (BMJ 2018;360:k80).

### Migraine, contraception and HRT

**Contraception:** Here's a quick overview (more in our article *Hormonal contraception and migraine*):

**WITH aura:** CHC: UKMEC 4 (UKMEC 3 if last migraine with aura  $\geq 5$ y ago). Other methods: UKMEC 2 or less.

**WITHOUT aura:** combined hormonal contraception: UKMEC 2 for initiation, UKMEC 3 if develops a new history of migraine while taking this method.

**Why?** CHC use in those with migraine is associated with 2–4x increased risk of ischaemic stroke (risk highest if aura).

**HRT:** can be used in those with all types of migraine. Use lowest oestrogen dose to control vasomotor symptoms.

(BMS):

- Post-hysterectomy: continuous transdermal oestrogen.

- Premenopause, with uterus: continuous transdermal oestrogen plus IUS.

- Postmenopause, with uterus: several options: 1. continuous transdermal oestrogen plus IUS or micronised progesterone. 2. continuous combined oestrogen and progestogen patches. 3. tibolone.